

Proton-coupled electron transfer drives the proton pump of cytochrome *c* oxidase

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Electron transfer in cell respiration is coupled to proton translocation across mitochondrial and bacterial membranes, which is a primary event of biological energy transduction. The resulting electrochemical proton gradient is used to power energy-requiring reactions, such as ATP synthesis. Cytochrome *c* oxidase is a key component of the respiratory chain, which harnesses dioxygen as a sink for electrons and links O₂ reduction to proton pumping¹. Electrons from cytochrome *c* are transferred sequentially to the O₂ reduction site of cytochrome *c* oxidase via two other metal centres, Cu_A and haem *a*, and this is coupled to vectorial proton transfer across the membrane by a hitherto unknown mechanism. On the basis of the kinetics of proton uptake and release on the two aqueous sides of the membrane, it was recently suggested that proton pumping by cytochrome *c* oxidase is not mechanistically coupled to internal electron transfer². Here we have monitored translocation of electrical charge equivalents as well as electron transfer within cytochrome *c* oxidase in real time. The results show that electron transfer from haem *a* to the O₂ reduction site initiates the proton pump mechanism by being kinetically linked to an internal vectorial proton transfer. This reaction drives the proton pump and occurs before relaxation steps in which protons are taken up from the aqueous space on one side of the membrane and released on the other².

Reduction of O₂ requires four electrons. One electron is transferred at a time to the binuclear haem *a*₃/Cu_B site of O₂ reduction, which therefore attains several intermediate states during the catalytic cycle. These states may be distinguished kinetically and their structures assigned by spectroscopic methods^{3–5}. O₂ reduction by cytochrome *c* oxidase is coupled to pumping of one proton across the membrane for each of the four transferred electrons¹. In addition, one 'substrate proton' per electron is taken up into the binuclear site from the negatively charged N-side of the membrane to form the product water. In the early stages of the catalytic cycle studied here, uptake of both substrate and pumped protons takes place via the D-pathway⁵, which is named after the residue D124, and ends at the residue E278 in the middle of the membrane (Fig. 1a; amino acid numbering corresponds to subunit I of cytochrome *c* oxidase from *Paracoccus denitrificans*).

Some structural features of cytochrome *c* oxidase are similar to those of the light-driven proton pump bacteriorhodopsin⁶. A recent study proposed a much closer similarity by concluding that the mechanism of proton pumping in cytochrome *c* oxidase is not kinetically coupled to electron transfer, but rather occurs as a result of conformational changes set forth by uptake of the substrate proton into the binuclear site². This issue is indeed best explored by studying the earliest part of the catalytic cycle² that immediately follows the reaction of the fully reduced enzyme with O₂ (Fig. 1b), because these reactions are unique in that electron transfer from haem *a* into the binuclear site (during A → P_R) is kinetically distinguishable from the next step (P_R → F)^{3–5,7} during which there is both net proton uptake

and pumping^{2,8,9}. These electron and proton transfers are kinetically indistinguishable in all later stages of the cycle. The kinetic separation may thus provide a fulcrum for a deeper general understanding of the proton pump mechanism, because it is reasonable to assume that the

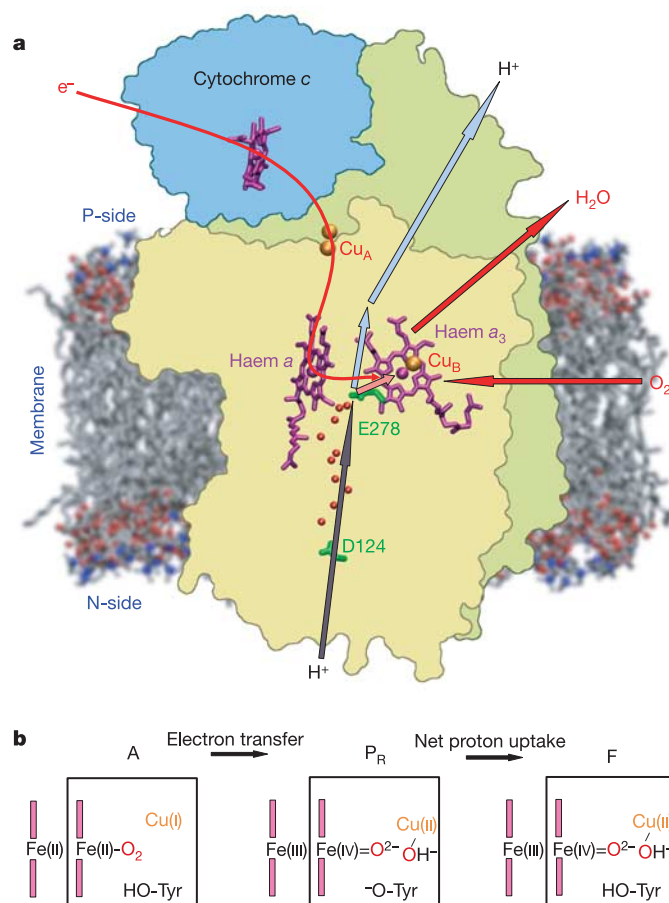


Figure 1 | Structure and function of cytochrome *c* oxidase. **a**, Subunits I (yellow) and II (green) are depicted in the phospholipid membrane, together with docked cytochrome *c* (blue, ref. 25). Proton transfer from the N-side of the membrane via the D-pathway (grey arrow) leads to E278. E278 donates protons, both to the haem *a*₃/Cu_B site to produce water from reduced O₂ (light red arrow), and for pumping across the membrane (blue arrows). The red spheres are crystallographically observed water molecules in the D-pathway (Protein Data Bank number 1v54, ref. 26). The VMD program²⁷ was used in producing the figure. **b**, Early intermediates in the reaction of fully reduced cytochrome *c* oxidase with O₂. The O₂ reduction site (square) includes haem *a*₃, Cu_B and a conserved tyrosine (Tyr)^{17,28,29}. Haem *a* is shown on the left of the square.

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same basic mechanism is repeated four times during the catalytic cycle.

In the early reaction sequence (Fig. 1b), the reduced binuclear site first binds an O_2 molecule ($\tau \approx 10 \mu s$) to form the oxygen adduct, compound A^{3-5} . Then, there is electron transfer from haem a into the binuclear site with scission of the O–O bond ($\tau \approx 30 \mu s$), leading to a ferryl/cupric intermediate that has been called P_R^7 . P_R is unstable and converts to the F state in the next step ($\tau \approx 80 \mu s$), which is linked to net uptake of a substrate proton from the N-side but involves no further electron transfer into the binuclear site^{2,5,10}. The earliest proton pumping event after O_2 binding has so far been ascribed to this latter reaction^{2,8,9} (but see below). The catalytic cycle continues from F to the oxidized O state ($\tau \approx 1-3$ ms) with transfer of a second electron into the binuclear site, pumping of one proton, and net uptake of another. Regeneration of the reduced binuclear site for binding the next O_2 molecule requires input of two more electrons, and these electron transfers are also each associated with net proton uptake¹¹ and pumping^{9,12}.

Elucidation of the mechanism of proton translocation may require monitoring of proton transfer within the enzyme structure in real time to complement measurements of proton uptake and release². Here we apply a sensitive electrometric technique⁸ to phospholipid vesicles inlaid with cytochrome c oxidase to monitor charge translocation within the membrane-bound enzyme. To determine its kinetic linkage to the catalytic chemistry, reaction intermediates are monitored by time-resolved optical spectroscopy. Two major kinetically distinguishable phases of charge translocation are observed at neutral pH (Fig. 2a) that roughly correspond to the spectroscopically observed kinetics of the $P_R \rightarrow F$ (Fig. 1b) and $F \rightarrow O$ transitions⁸. Each phase is mainly due to proton transfer from the aqueous N-side of the membrane into the binuclear site, and to proton pumping, with only minor contributions from electron transfer. Electron transfer from haem a into the binuclear site does not give rise to membrane potential due to the location of these centres at the same depth within the membrane¹³. The reaction $A \rightarrow P_R$ (Fig. 1b) has indeed been thought to be electrically silent⁸. However, the large amplitude of charge translocation during the subsequent $P_R \rightarrow F$ reaction might overwhelm possible charge translocation during $A \rightarrow P_R$. To explore this, the reaction was studied at high pH, where the $P_R \rightarrow F$ and $F \rightarrow O$ steps are known to be decelerated¹⁴. As shown in Fig. 2a, charge translocation during $F \rightarrow O$ is indeed slowed down. Paradoxically, however, the rate of the fast electrometric phase appears to be accelerated and the initial lag phase shortened (Fig. 2b). In fact, at high pH the fast phase coincides with the kinetics of the prior reaction step, $A \rightarrow P_R$, as monitored spectroscopically (Fig. 2c). Such an apparent acceleration can be explained if the fast phase of potential generation consists of two electrogenic events, the amplitudes of which are differently affected by pH: the amplitude during $A \rightarrow P_R$ is pH-independent, but during $P_R \rightarrow F$ the initially large amplitude decreases with pH. The transient during $A \rightarrow P_R$ is therefore very difficult to discern at neutral pH. Kinetic simulations indeed show that charge translocation during $A \rightarrow P_R$ is compatible with the data also at neutral pH (Fig. 2a, b, dashed lines), but independent evidence is obviously required. We therefore tested the D124N mutant enzyme where proton transfer to E278 via the D-pathway is strongly inhibited (Fig. 1a, ref. 15), and where proton pumping is abolished¹⁶. Therefore, the amplitude of the major electrometric phase is decreased, and the early transient during $A \rightarrow P_R$ should appear if it is present at neutral pH. As shown in Fig. 2b, charge translocation in this mutant enzyme is superimposable on the high pH trace of wild-type enzyme. Hence, an early phase of charge translocation indeed takes place during the $A \rightarrow P_R$ reaction also at neutral pH. Whether the electron transfer during $A \rightarrow P_R$ is essential can be assessed by studying the O_2 reaction of the 'mixed valence' enzyme, where only the binuclear site is initially reduced and no electron transfer from haem a is possible. Here, scission of the O–O bond occurs as before, but by reactions entirely

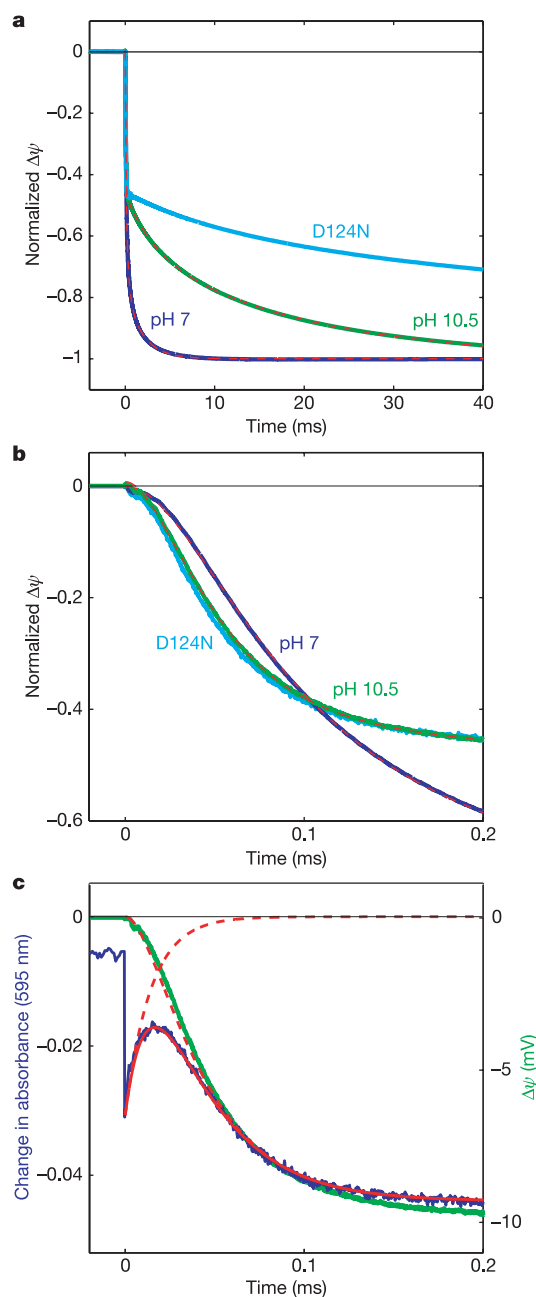


Figure 2 | Charge translocation and electron transfer during the reaction of fully reduced cytochrome c oxidase with O_2 . **a**, Normalized electrometric response of the reconstituted wild-type enzyme at pH 7 (blue), pH 10.5 (green) and with the D124N mutant enzyme at pH 7 (cyan). Time zero corresponds to the moment of the laser flash (see Methods). Red dashed lines on the pH 7 and 10.5 traces represent the theoretical fit of the data with a model of five sequential reaction steps⁸, where the rate constants for the formation and decay of compound A (Fig. 1b) were fixed at the values obtained from the optical data in **c**. **b**, The same traces as in **a** expanding the kinetics of the fast phase. **c**, Comparison of the time courses of formation and decay of compound A (Fig. 1b) at 595 nm (blue trace), with generation of electric potential (green trace), both at pH 10.5. The initial downward deflection of the blue trace is due to CO photo-dissociation. The solid red trace is a two-component exponential fit to the optical data at 595 nm; the dashed red lines show the two components, of which the second (decay of compound A) coincides with the electrometric trace. $\Delta\psi$, change in membrane potential.

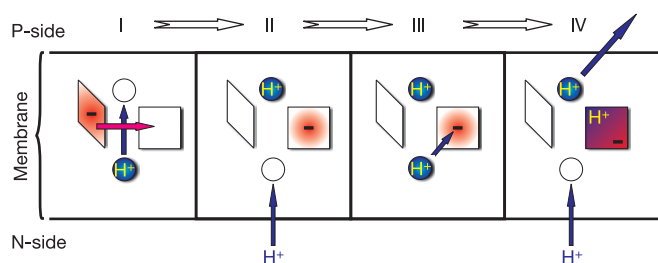


Figure 3 | Scheme of the proposed proton pump mechanism. Four states are shown (I–IV), each comprising haem *a* (rhombus) and the haem *a*₃/Cu_B site (square). The lower and upper circles denote the carboxylic residue E278 at the end of the D-pathway (Fig. 1a) and an unidentified protonatable site above the haems, respectively. In I → II electron transfer from haem *a* to the binuclear site is coupled to transfer of a proton from E278 to the protonatable site. In II → III, E278 is re-protonated from the N-side via the D-pathway. In III → IV, a substrate proton is transferred from E278 to the binuclear site. After IV, E278 is again re-protonated and the proton above the haems is ejected towards the P-side.

within the binuclear site¹⁷. As shown previously⁸, this reaction is not associated with any charge translocation. We therefore conclude that the early charge translocation during A → P_R not only coincides kinetically with electron transfer from haem *a* to the binuclear site, but also depends on it.

At neutral pH the amplitude of charge translocation during formation of the F state corresponds to translocation of about 1.8 charge equivalents across the membrane¹⁸. Vectorial charge movement during A → P_R (approximately 15% of the amplitude at neutral pH) thus amounts to transfer of one electrical charge across about 30% of the membrane dielectric. Because it cannot be due to transmembrane electron transfer, it probably reflects an internal proton transfer reaction towards the positively charged P-side. The residue E278 at the end of the D-pathway is normally protonated before the reaction with O₂. This residue is involved in proton pumping¹⁹ as well as in proton transfer to the binuclear site¹⁵, and is thus a candidate for being the proton donor. This is supported by electrometric experiments with the E278Q mutant enzyme, where virtually no charge translocation corresponding to the A → P_R reaction was observed (see Supplementary Information), although electron transfer still occurs from haem *a* to the binuclear site¹⁵. The proton acceptor cannot be the binuclear site because proton uptake into this site would produce intermediate F, which is formed only in the next reaction. We suggest, therefore, that the charge translocation during A → P_R is due to internal transfer of a proton from E278 to a so far unidentified location above the haem groups^{2,14,20–22} (I → II, Fig. 3), which is the initial step of the proton pump mechanism. It is followed by protonation of E278 via the D-pathway (II → III), proton transfer to the binuclear site (III → IV), re-protonation of E278 and release of the pumped proton to the aqueous P-side. The latter reactions all occur secondarily during the P_R → F transition, which accounts for the fact that most of the charge translocation amplitude is kinetically linked to this reaction.

Our results are consistent with the recent observation² that proton release and uptake on the two aqueous sides of the membrane are kinetically coupled to the P_R → F reaction. However, we show here that these events are preceded by an internal vectorial proton transfer reaction that is kinetically associated with electron transfer from haem *a* to the O₂ reduction site, and which can be ascribed as the initial driving step of the proton pump mechanism (Fig. 1a, short blue arrow; Fig. 3, I → II). This finding has its parallel in the proton pump of bacteriorhodopsin, where mechanistically important internal proton transfers precede proton equilibration with the surroundings^{6,23}. Important details of the mechanism still remain unsolved, however, such as the identity of the proton-binding site above the haem groups, and the reason for why proton transfer from

E278 to that site occurs before transfer of the substrate proton to the binuclear site^{21,22,24}.

METHODS

Enzyme preparation and reconstitution into phospholipid vesicles. Bacterial growth, isolation of bacterial membranes and purification of cytochrome *c* oxidase from *P. denitrificans*, as well as reconstitution of wild-type and mutant enzymes into proteoliposomes, were done by described methods⁸, except that the enzyme concentration during reconstitution was increased to 6 μM.

Electrometric measurements. A detailed description of the electrometric method has been published⁸. The voltage recorded is proportional to the extent of charge translocation perpendicular to the membrane plane. The pH dependence of the electrometric response was measured in a variety of buffers: 100 mM MES (in the range pH 6.0–6.6), MOPS (pH 6.6–7.8), Tris (pH 7.8–8.8), Bis-Tris propane (pH 8.8–9.2), CHES (pH 9.2–9.8) and CAPS (pH 9.8–10.5). Media were supplemented with 50 mM glucose, 0.3 mg ml⁻¹ catalase, 3 mg ml⁻¹ glucose oxidase and 1 μM hexaammine ruthenium (III).

Transient optical spectroscopy. The reaction of the solubilized fully reduced wild-type enzyme with oxygen was measured on a microsecond to millisecond timescale. About 30 μM enzyme in 100 mM CAPS, pH 10.5, 0.02% dodecyl maltoside, was made anaerobic on a vacuum line, reduced with 20 mM potassium ascorbate and 5 μM TMPD (*N,N,N',N'*-tetramethyl-*p*-phenylenediamine) and incubated with 100% CO. The sample was loaded into one syringe of a stopped-flow system (RX2000, Rapid Kinetics Spectrometer Accessory, Applied Photophysics) and mixed with a fivefold volume of oxygen-saturated buffer. The reaction was started by a laser flash to photo-dissociate CO bound to the enzyme.

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- Wikström, M. K. F. Proton pump coupled to cytochrome *c* oxidase in mitochondria. *Nature* **266**, 271–273 (1977).
- Faxén, K., Gilderson, G., Ådelroth, P. & Brzezinski, P. A mechanistic principle for proton pumping by cytochrome *c* oxidase. *Nature* **437**, 286–289 (2005).
- Babcock, G. T. & Wikström, M. O₂ activation and the conservation of energy in cell respiration. *Nature* **356**, 301–309 (1992).
- Ferguson-Miller, S. & Babcock, G. T. Heme/copper terminal oxidases. *Chem. Rev.* **96**, 2889–2907 (1996).
- Gennis, R. B. Coupled proton and electron transfer reactions in cytochrome oxidase. *Front. Biosci.* **9**, 581–591 (2004).
- Wikström, M. Proton translocation by bacteriorhodopsin and heme-copper oxidases. *Curr. Opin. Struct. Biol.* **8**, 480–488 (1998).
- Morgan, J. E., Verkhovskiy, M. I., Palmer, G. & Wikström, M. The role of the P_R intermediate in the reaction of cytochrome *c* oxidase with O₂. *Biochemistry* **40**, 6882–6892 (2001).
- Jasaitis, A., Verkhovskaya, M., Morgan, J. E., Verkhovskiy, M. & Wikström, M. Assignment and charge translocation stoichiometries of the major electrogenic phases in the reaction of cytochrome *c* oxidase with dioxygen. *Biochemistry* **38**, 2697–2706 (1999).
- Bloch, D. et al. The catalytic cycle of cytochrome *c* oxidase is not the sum of its two halves. *Proc. Natl Acad. Sci. USA* **101**, 529–533 (2004).
- Namslauer, A., Aagaard, A., Katsonouri, A. & Brzezinski, P. Intramolecular proton transfer reactions in a membrane-bound proton pump: the effect of pH on the peroxy to ferryl transition in cytochrome *c* oxidase. *Biochemistry* **42**, 1488–1498 (2003).
- Mitchell, R. & Rich, P. R. Proton uptake by cytochrome *c* oxidase on reduction and on ligand binding. *Biochim. Biophys. Acta* **1186**, 19–26 (1994).
- Ruitenbergh, M., Kannt, A., Bamberg, E., Fendler, K. & Michel, H. Reduction of cytochrome *c* oxidase by a second electron leads to proton translocation. *Nature* **417**, 99–102 (2002).
- Iwata, S., Ostermeier, C., Ludwig, B. & Michel, H. Structure at 2.8 Å resolution of cytochrome *c* oxidase from *Paracoccus denitrificans*. *Nature* **376**, 660–669 (1995).
- Brändén, G. et al. The protonation state of a heme propionate controls electron transfer in cytochrome *c* oxidase. *Biochemistry* **44**, 10466–10474 (2005).
- Ådelroth, P., Svensson-Ek, M., Mitchell, D. M., Gennis, R. B. & Brzezinski, P. Glutamate 286 in cytochrome *aa*₃ from *Rhodospirillum rubrum* is involved in proton uptake during the reaction of the fully reduced enzyme with dioxygen. *Biochemistry* **36**, 13824–13829 (1997).
- Thomas, J. W., Puustinen, A., Alben, J. O., Gennis, R. B. & Wikström, M. Substitution of asparagine for aspartate 135 in subunit I of the cytochrome *bo* ubiquinol oxidase of *Escherichia coli* eliminates proton pumping activity. *Biochemistry* **32**, 10923–10928 (1993).
- Babcock, G. T. How oxygen is activated and reduced in respiration. *Proc. Natl Acad. Sci. USA* **96**, 12971–12973 (1999).
- Verkhovskiy, M. I., Jasaitis, A., Verkhovskaya, M. L., Morgan, J. E. & Wikström, M. Proton translocation by cytochrome *c* oxidase. *Nature* **400**, 480–483 (1999).

19. Verkhovskaya, M. L. *et al.* Glutamic acid 286 in subunit I of cytochrome *bo*₃ is involved in proton translocation. *Proc. Natl Acad. Sci. USA* **94**, 10128–10131 (1997).
20. Michel, H. Cytochrome *c* oxidase: catalytic cycle and mechanism of proton pumping - a discussion. *Biochemistry* **38**, 15129–15140 (1999).
21. Wikström, M., Verkhovsky, M. I. & Hummer, G. Water-gated mechanism of proton translocation by cytochrome *c* oxidase. *Biochim. Biophys. Acta* **1604**, 61–65 (2003).
22. Popovic, D. M. & Stuchebrukhov, A. A. Proton pumping mechanism and catalytic cycle of cytochrome *c* oxidase: Coulomb pump model with kinetic gating. *FEBS Lett.* **566**, 126–130 (2004).
23. Lanyi, J. K. & Luecke, H. Bacteriorhodopsin. *Curr. Opin. Struct. Biol.* **11**, 415–419 (2001).
24. Siegbahn, P. E. M., Blomberg, M. R. A. & Blomberg, M. L. Theoretical study of the energetics of proton pumping and oxygen reduction in cytochrome oxidase. *J. Phys. Chem. B* **107**, 10946–10955 (2003).
25. Flöck, D. & Helms, V. Protein-protein docking of electron transfer complexes: cytochrome *c* oxidase and cytochrome *c*. *Proteins* **47**, 75–85 (2002).
26. Tsukihara, T. *et al.* The low spin heme of cytochrome *c* oxidase as the driving element of the proton-pumping process. *Proc. Natl Acad. Sci. USA* **100**, 15304–15309 (2003).
27. Humphrey, W., Dalke, A. & Schulten, K. VMD: Visual molecular dynamics. *J. Mol. Graph.* **14**, 33–38 (1996).
28. Yoshikawa, S. *et al.* Redox-coupled crystal structural changes in bovine heart cytochrome *c* oxidase. *Science* **280**, 1723–1729 (1998).
29. Buse, G., Soulimane, T., Dewor, M., Meyer, H. E. & Bluggel, M. Evidence for a copper-coordinated histidine-tyrosine cross-link in the active site of cytochrome oxidase. *Protein Sci.* **8**, 985–990 (1999).

Supplementary Information is linked to the online version of the paper at www.nature.com/nature.

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